SafeTrip AI

AI-Powered Travel Planning & Tourist Safety Platform

Product Requirements, UI/UX Blueprint, Technical Architecture & Antigravity Build Guide

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1. Project Overview

Project Name

SafeTrip AI

Tagline

Plan Smart. Travel Safer.

One-Line Vision

SafeTrip AI is an AI-powered travel planning and tourist-safety platform that helps users create personalized trips, get smart route recommendations, receive contextual safety alerts, share live location during active trips, and use SOS support in emergencies.

Core Problem

Travelers often face problems such as:

- They do not know where to go, what to explore, or how much a trip will cost.
- They struggle to choose the best route according to traffic, road conditions, travel time, and safety.
- Solo travelers and families may feel unsafe in unfamiliar locations.
- Network can become weak in remote, mountain, forest, or rural areas.
- Users may enter unsafe, restricted, isolated, or accident-prone areas unknowingly.
- In an emergency, it is difficult to quickly communicate a user's exact or last known location.
- Families/admins may not know whether the traveler is safe, offline, or in a danger area.

Solution

SafeTrip AI combines:

1. **AI Trip Planner**
2. **Personalized Itinerary Generator**
3. **Route and Traffic Intelligence**
4. **Live Travel Safety Dashboard**
5. **GPS-Based Zone Monitoring**
6. **Network and Last-Seen Monitoring**
7. **SOS Emergency Alert System**
8. **Admin Safety Operations Dashboard**
9. **AI Assistant for Travel and Safety Queries**

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2. Product Objectives

Primary Objectives

- Help users create complete travel plans based on destination, budget, dates, duration, interests, and group size.
- Provide route recommendations based on traffic, estimated duration, road closures, and available safety information.
- Offer consent-based live tracking during an active trip.
- Detect safety risk using location zone, network quality, route deviation, time context, and GPS freshness.
- Send meaningful alerts when a user approaches a caution or danger zone.
- Save the user's last known location if their signal or internet connection is lost.
- Allow users to trigger SOS and notify the admin safety team plus selected emergency contacts.
- Provide an admin dashboard for monitoring high-priority cases and responding quickly.
- Use AI to make the product more personalized and helpful, while keeping critical safety decisions controlled by verified rules and humans.

Important Product Boundary

SafeTrip AI should never claim:

- "You are completely safe."
- "This route is guaranteed safe."
- "Police/ambulance has been dispatched."

Unless there is an officially verified and tested emergency-service integration, the correct wording should be:

> “Your SOS alert has been sent to the SafeTrip safety team and your selected emergency contacts.”

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3. User Types

User Type	Description	Main Needs
Traveler	A person planning or currently taking a trip	Trip planning, itinerary, map, safety alerts, SOS
Solo Traveler	A user traveling alone	Safer routes, live safety status, emergency contacts
Family Traveler	A user traveling with family or children	Comfortable routes, nearby hospitals, family-friendly plans
Adventure Traveler	A user visiting mountains, treks, forests, remote areas	Weather alerts, weak-network alerts, route tracking
Admin	Platform safety operator	Live alerts, map view, SOS handling, case notes
Super Admin	System administrator	User management, zones, policies, analytics, permissions

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4. Complete User Flow

Step 1: Landing Page

When a user opens the website, they see:

- Tourist place animated/random background
- Logo: **SafeTrip AI**
- Tagline: **Plan Smart. Travel Safer.**
- Buttons:
    - Login

    - Register
    - Explore Destinations
- Small safety message:
    - “Travel guidance and safety support during your active trip.”

Design Direction

- Full-screen destination image/video background.
- Background changes between mountains, beaches, cities, historical places, forests, and deserts.
- Dark transparent overlay so text remains readable.
- Premium travel look with calm colors, not an alarming security-app look.

Step 2: Register

A new user should register before using the platform.

Required Fields

- Full name
- Email address
- Phone number
- Password or OTP-based authentication
- Date of birth — optional
- Gender — optional, only if useful for personalized safety preferences
- City/country — optional
- Emergency contact name
- Emergency contact phone number
- Relationship with emergency contact

Required Consent Checkboxes

- I agree to the Terms and Privacy Policy.
- I allow SafeTrip AI to use my location only during an active trip safety session.
- I understand that location tracking can be paused or stopped by me.
- I allow SOS alerts to be shared with my selected emergency contacts.

Important Rule

Do **not** request location permission during registration. Ask for it only when the user starts an active trip and understands why location access is needed.

Step 3: Login

Existing users log in using:

- Email + password
- Phone + OTP
- Google login — optional
- Apple login — optional

After successful login:

- Show destination-themed loading screen.
- Display:
  “Hello, `<User Name>`! Ready for your next safe journey?”

Then redirect user to the Home Dashboard.

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5. Home Dashboard

The home dashboard should feel exciting, visual, and travel-focused.

Main Sections

Welcome Section

Example:

> Hello, Rohan!
> Where would you like to travel next?

User Statistics Cards

- Total Trips Planned
- Completed Trips
- Active Trip
- Estimated Travel Spend
- Saved Destinations
- Active Travelers Today — only show this if the metric is real and clearly defined

Travel Gallery

Show destination photos such as:

- Goa
- Manali
- Jaipur
- Kashmir
- Kerala
- Ladakh
- Varanasi
- Rishikesh
- Mumbai
- Delhi
- Meghalaya

Main CTA

A large button:

> **Make a Trip**

Secondary Buttons

- My Trips
- Safety Dashboard
- Saved Places
- Emergency Contacts
- Profile

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6. Make a Trip Flow

This should be a multi-step wizard, not one large confusing form.

Step 1: Destination

Ask:

> Where do you want to go?

Fields:

- Starting location
- Destination
- Multiple destinations — optional
- City, state, country
- Destination suggestions/autocomplete

Example:

- From: Delhi
- To: Manali

Step 2: Dates and Duration

Ask:

> When are you planning to travel?

Fields:

- Start date
- End date
- Number of days
- Flexible dates toggle
- Preferred departure time

Example:

- Start date: 10 December 2026
- End date: 15 December 2026
- Duration: 5 days

Step 3: Budget

Ask:

> What is your estimated budget?

Fields:

- Total budget
- Budget per person

- Low / Medium / Premium
- Currency
- Travel group size

Example:

- ₹25,000 per person
- 2 travelers
- Mid-range trip

Step 4: Travel Interests

Ask:

> What do you want to explore?

Options:

- Adventure
- Trekking
- Mountains
- Beaches
- Religious places
- Food
- Shopping
- Historical places
- Nature
- Wildlife
- Photography
- Nightlife
- Luxury stay
- Relaxation
- Local culture

Step 5: Travel Preferences

Fields:

- Transport preference:
    - Flight

    - Train
    - Bus
    - Car
    - Bike
    - Public transport
- Stay preference:
    - Hostel
    - Budget hotel
    - Mid-range hotel
    - Resort
    - Homestay
- Food preference:
    - Vegetarian
    - Non-vegetarian
    - Vegan
    - Any
- Accessibility needs — optional
- Traveling with children/senior citizens — optional

Step 6: AI Trip Plan Result

After user submits details, SafeTrip AI generates:

- Unique Trip ID
- Destination-specific theme
- Day-wise itinerary
- Approximate budget split
- Travel route options
- Estimated travel duration
- Best time to leave
- Hotels/stay suggestions
- Tourist places
- Food recommendations
- Packing checklist
- Safety tips for the destination
- Emergency numbers and nearby help locations where available

Example Trip ID:

TRP-IND-2026-000827

This Trip ID should be visible to:

- User
- Admin dashboard
- SOS record
- Support team
- Emergency workflow

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7. Destination Theme Customization

The website theme should update based on destination type.

Destination Type	Theme Style
Goa / Beach	Blue ocean, turquoise gradients, palm visuals
Manali / Kashmir	Snow mountains, cool blue, white, fog effects
Rajasthan	Sand, royal gold, palace patterns
Kerala	Green nature, rivers, tropical gradients
Ladakh	Brown mountains, starry sky, rugged roads
Cities	Modern blue/purple city lights
Religious trips	Calm saffron, white, peaceful imagery
Forest/wildlife	Deep green, natural earthy colors

UI Rule

Theme should enhance the experience but must not reduce readability or accessibility.

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8. Active Trip Safety Dashboard

When the trip starts, the user should have access to the **Active Trip Safety Dashboard**.

Top Navigation

Show:

- SafeTrip AI logo
- Welcome, `<User Name>`
- Current Trip ID
- Current Status badge
- Notification icon
- Profile icon

Example:

> Welcome, Rohan
> Trip ID: TRP-IND-2026-000827
> Status: CAUTION

Four Main Tabs

Tab 1: User Information

Show:

- User ID
- Trip ID
- User name
- Email
- Phone number
- Emergency contact
- Destination
- Travel dates
- Current transport mode
- Tracking status

Example:

User: Rohan Sharma
Trip ID: TRP-IND-2026-000827
Email: rohan@example.com
Phone: +91 XXXXX XXXXX
Emergency Contact: Amit Sharma
Tracking: Active

Tab 2: Safety Status

This is the main safety card.

Show:

- Overall safety status
- Area risk status
- Network status
- GPS accuracy
- Last location update
- Route deviation
- Current warning
- Recommended action

Possible statuses:

- Safe
- Caution
- Danger
- Offline / Last Seen

Example:

Overall Status: CAUTION
Area: Yellow Zone
Network: Weak
GPS Update: 22 seconds ago
Route Deviation: 350 meters
Recommendation: Stay on the suggested main route.

Tab 3: Live Map

Map should display:

- User’s current GPS position
- Last known position if offline
- Planned route
- Alternative routes
- Safe zone overlays
- Caution zone overlays
- Danger zone overlays

- Nearby police station
- Nearby hospital
- Nearby petrol pump
- Nearby hotel
- Nearby transport station
- Current traffic information
- Road closures when provider data is available

Map Colors

Zone	Color	Meaning
Safe Zone	Green	Lower-risk/populated/verified safer area
Caution Zone	Yellow / Amber	Stay alert; limited facilities or weak network possible
Danger Zone	Red	Restricted, isolated, high-risk, unsafe or incident-prone area
Offline / Last Seen	Grey	Location has not updated recently

Tab 4: SOS

This tab must be very clear and easy to use.

Show:

- Large red SOS button
- Emergency contact details
- Current/last known location
- Safety team alert status
- Call emergency contact button
- Share live location button
- Cancel countdown button

SOS Flow

1. User presses SOS.
2. App shows a 5-second countdown.
3. User can cancel accidental activation.
4. If not cancelled:
    - SOS event is created.
    - Current GPS location is collected.

    - Last 5–10 location points are attached.
    - User ID and Trip ID are attached.
    - Admin gets high-priority notification.
    - Emergency contacts get message/push/SMS according to consent.
    - User sees confirmation screen.

Example confirmation:

> SOS activated.
> Your location and trip details have been sent to the SafeTrip safety team and your selected emergency contacts.

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9. Safety Zone Logic

Important Principle

Do not decide user safety only from network strength.

Weak network does not always mean danger. A mountain, forest, or remote tourist destination may have poor network even if the area is not unsafe.

Therefore, separate these signals:

1. **Area Risk**
2. **Network Condition**
3. **Route Deviation**
4. **Time of Day**
5. **GPS Freshness and Accuracy**
6. **Traffic / Road / Weather Events**
7. **User Movement Pattern**

Zone Types

Safe Zone

Examples:

- Popular tourist locations

- Main city roads
- Hotels
- Markets
- Transport hubs
- Police station surroundings
- Hospitals
- High-footfall locations

User message:

> You are currently in a safer, active area. Continue following your planned route.

Caution Zone

Examples:

- Weak-network areas
- Low-footfall roads
- Remote scenic routes
- Late-night travel areas
- Poorly lit roads
- Areas near route deviation
- Places with limited facilities

User message:

> You are entering a caution area. Please stay alert, keep your device charged, and remain close to your planned route.

Danger Zone

Examples:

- Restricted zones
- No-entry areas
- Accident-prone roads
- Isolated risky roads
- High-risk areas verified by trusted sources
- Forest boundaries
- Dangerous terrain
- Unofficial/unsafe routes
- Areas under weather or road-closure warning

User message:

> Warning: You are near or inside a higher-risk area. Avoid proceeding further unless necessary. Use the recommended route or contact support if you need help.

Safety Risk Score

Start with transparent rule-based logic.

Risk Score =
0.35 × Area Risk
+ 0.20 × Route Deviation
+ 0.15 × Time Risk
+ 0.10 × Network Risk
+ 0.10 × GPS Quality Risk
+ 0.10 × Traffic / Weather / Road Risk

Risk State Table

Risk Score	Status	Meaning
0–34	Safe	Low risk based on current available signals
35–64	Caution	User should remain alert
65–100	Danger	High-risk combination detected
Not applicable	Offline / Last Seen	Location updates are stale or unavailable

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Example Risk Scenarios

Situation	Status
User is on main route in a populated tourist area with recent GPS update	Safe
User is in a known safe area but network is weak	Caution, not Danger
User is 2 km away from planned route at 11:30 PM	Caution or Danger depending on area

User enters a red zone with weak network and route deviation	Danger
User location stops updating near a high-risk area	High-priority admin review
User goes offline in a safe hotel area	Offline / Last Seen, not automatic danger
User manually presses SOS	Critical regardless of score

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10. Live GPS Tracking Logic

Tracking Consent

Tracking should start only after the user explicitly chooses:

> Start Trip Safety Tracking

The consent modal must tell the user:

- Why location is collected
- When tracking starts
- When tracking stops
- Who can see the data
- How to pause tracking
- How long location history is retained
- What happens during SOS

Location Update Frequency

Do not send GPS data every second in normal conditions.

Situation	Suggested Update Frequency
User is moving normally	Every 20–30 seconds
User is stationary	Every 60–120 seconds
Weak network	Every 60 seconds with retry queue

Caution zone	Every 15–20 seconds
Danger zone	Every 10 seconds if technically possible
SOS active	Every 5–10 seconds if possible
Low battery	Reduce frequency and inform user

Offline and Signal Loss Logic

If user loses network:

1. Save the latest GPS location locally.
2. Store timestamp and accuracy.
3. Show:
   Last updated 4 minutes ago.
4. Display user as:
   Offline / Last Seen
5. When connection returns:
   - Sync queued location records.
   - Mark user active again.
   - Update admin dashboard.

Important Rule

Do not falsely claim the user is in danger only because internet is lost.

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11. Route Intelligence

Route Features

The user should receive:

- Fastest route
- Lower-risk route
- Scenic route
- Budget-friendly route
- Traffic-aware ETA

- Alternative route
- Road-closure notice where available
- Accident/incident advisory where available
- Route deviation detection

Route Recommendation Table Example

Route	ETA	Traffic	Network Coverage	Safety Indicator	Recommendation
Route A	4h 20m	Medium	Good	Medium	Fastest route
Route B	4h 55m	Low	Better	High	Recommended lower-risk route
Route C	5h 15m	Low	Weak in parts	Medium	Scenic daytime route

Correct Language

Use:

- “Recommended based on available traffic and area data”
- “Lower-risk route”
- “Better network coverage route”
- “Avoid after dark”
- “Traffic-aware ETA”

Do not use:

- “100% safe route”
- “Guaranteed accident-free route”

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12. AI Features

AI Feature 1: Natural Language Trip Planner

User can write in English, Hindi, or Hinglish:

> Mujhe Delhi se Manali 5 din ke liye jana hai. Budget 25,000 hai aur mujhe snowfall, adventure aur trekking pasand hai.

AI extracts:

```
{
  "source": "Delhi",
  "destination": "Manali",
  "days": 5,
  "budget": 25000,
  "interests": [
    "snowfall",
    "adventure",
    "trekking"
  ],
  "travelers": 1
}
```

The system should ask follow-up questions if information is missing.

AI Feature 2: Personalized Itinerary

AI can create:

- Day-wise plan
- Budget allocation
- Activity suggestions
- Food recommendations
- Packing list
- Local travel suggestions
- Destination safety advice
- Hotel-area preference suggestions
- Best travel time suggestions

Example

For a family with children:

- Short travel distances
- Child-friendly places
- Hospitals nearby
- Comfortable stay options
- Daytime activities

For a solo traveler:

- Populated stay areas
- Daytime arrival suggestions
- Main-route travel
- Emergency contact setup
- Location-sharing reminders

AI Feature 3: Safety Alert Explanation

The rule engine decides the risk state. AI explains it in clear language.

Example:

> You are 1.8 km away from your planned route. The current area has weak connectivity, and it is late evening. Please return to the main route or choose the recommended alternative route.

AI Feature 4: Admin Incident Summary

When SOS or danger event occurs, AI summarizes the situation.

Example:

> User: Rohan Sharma
> Trip ID: TRP-IND-2026-000827
> SOS triggered at 10:34 PM.
> Last confirmed location: XYZ Road, 2 minutes ago.
> User entered a high-risk zone 8 minutes earlier.
> Network changed from 4G to offline.
> Route deviation: 1.4 km.

Admin can then take action faster.

AI Feature 5: Travel Chatbot

User may ask:

- “Mere budget mein kya best hai?”
- “Manali mein 3 din ka plan bana do.”
- “Nearby hospital kahan hai?”
- “Meri route par traffic hai kya?”
- “Signal weak ho raha hai, mujhe kya karna chahiye?”
- “Main safe area mein hoon?”
- “SOS kaise activate karun?”

Important AI Rule

AI should use controlled backend tools/functions:

getTripDetails(tripId)
getCurrentSafetyState(tripId)
getCurrentRoute(tripId)
getTraffic(routeId)
getNearbyHospital(latitude, longitude)
getNearbyPolice(latitude, longitude)
getEmergencyContacts(userId)
createSOSDraft(userId)

AI should not get unrestricted access to all raw location history.

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13. What AI Must Not Do

Feature	Should AI Handle It Alone?	Correct Approach
Login/register	No	Standard secure backend logic
User ID generation	No	Database-generated ID
GPS collection	No	Geolocation/native device APIs
SOS trigger	No	Deterministic emergency workflow
Sending SOS automatically	Not alone	User action or defined escalation policy
Traffic truth	No	Maps/traffic provider is source of truth
Accident truth	No	Official/verified data sources

Area danger truth	Not alone	Verified zone database + rules + human review
Safety explanation	Yes, with guardrails	AI uses structured event data
Itinerary planning	Yes	AI + maps/places/hotel data
Admin summaries	Yes	AI summarizes verified inputs
Final emergency response	No	Human admin/team decision

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14. Recommended Technology Stack

Frontend

Technology	Use
Next.js	Main frontend framework
React	UI components
TypeScript	Type safety
Tailwind CSS	Responsive styling
shadcn/ui	Reusable UI components
Framer Motion	Smooth animations
React Hook Form	Forms and validation
Zod	Schema validation
TanStack Query	API data fetching/cache
Zustand or Redux Toolkit	App state management
PWA support	Installable mobile-like website

Backend

Technology	Use
NestJS with Node.js	Recommended backend if using TypeScript everywhere
FastAPI with Python	Alternative if AI/ML work is Python-heavy
REST API	Standard client-server operations
WebSockets / Socket.IO	Live location and alert updates
BullMQ	Background jobs
Redis	Cache, queues, rate limits, live state

Database

Technology	Use
PostgreSQL	Main relational database
PostGIS	Geographic zones, coordinates, route queries
Prisma	ORM for Node.js/NestJS
SQLAlchemy	ORM alternative for FastAPI/Python

Maps and Location

Technology	Use
Google Maps Platform	Maps, routes, traffic, places, geocoding
Mapbox	Alternative maps/routing provider
Leaflet	Map UI library
OpenStreetMap	Lower-cost/open-source map layer
Browser Geolocation API	Web GPS location access

Native Android/iOS location APIs	Future reliable background tracking

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Notifications

Technology	Use
Firebase Cloud Messaging	Mobile/push notifications
Web Push	Browser notifications
Twilio / MSG91	SMS notifications
Email provider	Email alerts and verification

AI Layer

Technology	Use
LLM API	Chatbot, itinerary, summaries, multilingual support
Function/tool calling	Controlled access to data/services
Structured JSON output	Reliable machine-readable AI responses
Vector database — optional	Destination guides, FAQs, policies
ML model — later	Anomaly detection and risk prioritization

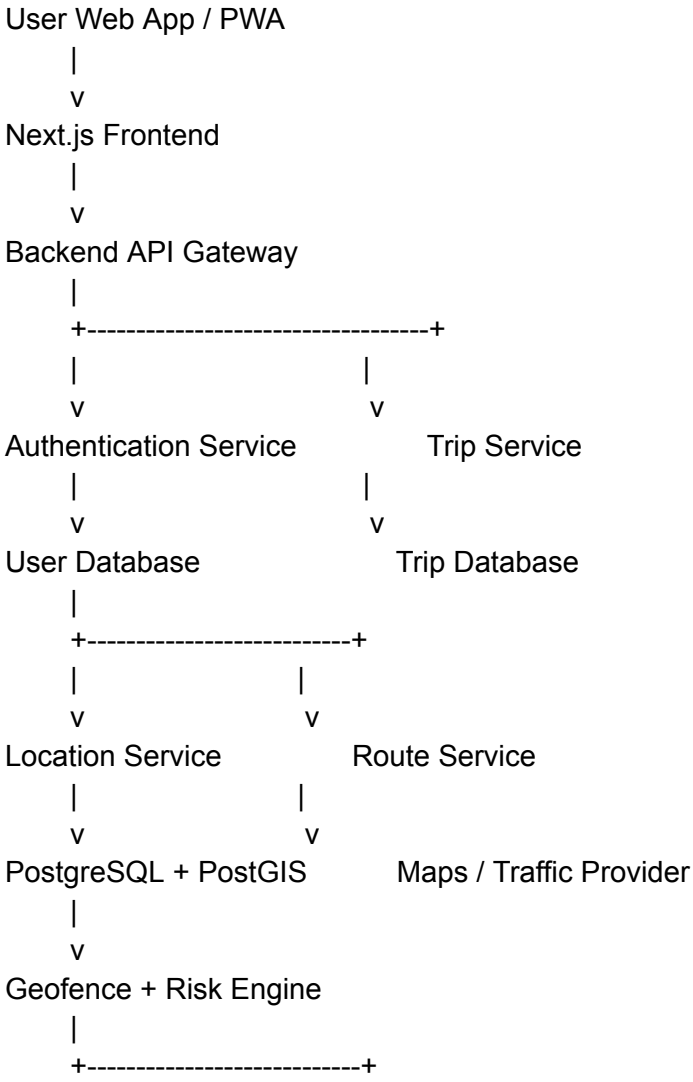
Security and Monitoring

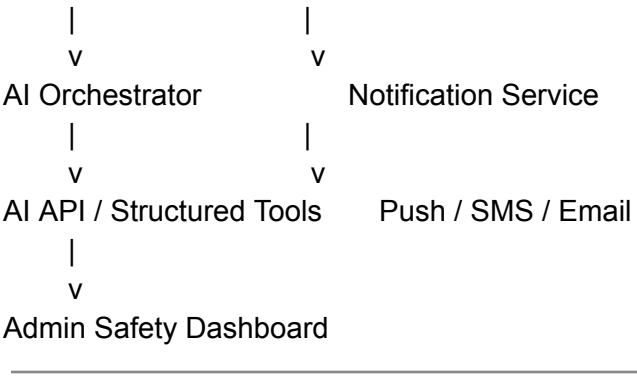
Technology	Use
HTTPS	Secure communication
Argon2 or bcrypt	Password hashing
JWT + refresh tokens or secure sessions	Authentication
RBAC	User/admin permissions

MFA for admins	Admin account protection
Sentry	Error monitoring
OpenTelemetry	Tracing
Grafana / Datadog	Monitoring and dashboards
AWS Secrets Manager / environment secrets	API key protection

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15. System Architecture





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16. Database Design

users

id
name
email
phone
password_hash
status
created_at
updated_at

user_consents

id
user_id
consent_type
consent_version
accepted_at
revoked_at

emergency_contacts

id
user_id
name
phone
relationship
is_verified
created_at

trips

id
trip_code
user_id
source_location
destination
start_date
end_date
days
budget
traveler_count
interests
transport_preference
stay_preference
status
created_at

trip_routes

id
trip_id
route_name
provider
encoded_polyline
distance_meters
duration_seconds
traffic_data
network_coverage_score
risk_score
updated_at

safety_zones

id
name
zone_type
risk_level
geometry_polygon
description
source
verified_by
verified_at
is_active

created_at

location_events

id
trip_id
user_id
latitude
longitude
accuracy_meters
altitude
speed
heading
network_status
battery_level
recorded_at
received_at

risk_events

id
trip_id
user_id
risk_score
risk_level
reason_codes
location_event_id
created_at

sos_alerts

id
trip_id
user_id
status
triggered_at
latest_location_event_id
acknowledged_by
acknowledged_at
resolved_at

admin_actions

id

admin_id
entity_type
entity_id
action
notes
created_at

notifications

id
user_id
trip_id
type
channel
payload
delivery_status
created_at

---

17. Core API Endpoints

Authentication

POST /auth/register
POST /auth/login
POST /auth/logout
POST /auth/verify-phone
POST /auth/forgot-password
POST /auth/reset-password
GET /auth/me

User Profile

GET   /users/me
PATCH /users/me
POST   /users/me/emergency-contacts
PATCH /users/me/emergency-contacts/:id
DELETE /users/me/emergency-contacts/:id

Trip

POST   /trips

GET /trips
GET /trips/:tripId
PATCH /trips/:tripId
DELETE /trips/:tripId
POST /trips/:tripId/ai-plan
GET /trips/:tripId/routes

Tracking and Safety

POST /trips/:tripId/tracking/start
POST /trips/:tripId/tracking/stop
POST /trips/:tripId/location-events
GET /trips/:tripId/safety-state
GET /trips/:tripId/location-history
GET /trips/:tripId/alerts

SOS

POST /sos
GET /sos/:sosId
POST /sos/:sosId/cancel

Admin

POST /admin/login
GET /admin/alerts
GET /admin/sos-alerts
GET /admin/users/:userId
GET /admin/trips/:tripId
POST /admin/sos/:sosId/acknowledge
POST /admin/sos/:sosId/resolve
POST /admin/zones
PATCH /admin/zones/:zoneId
DELETE /admin/zones/:zoneId

---

18. WebSocket Events

location.updated
location.offline
safety.status.changed
risk.event.created
alert.created

route.updated
sos.triggered
sos.acknowledged
sos.resolved
notification.sent
notification.delivered

---

19. Admin Dashboard

Admin Dashboard Sections

1. Alert Queue

Show priority alerts:

- SOS active
- Danger-zone entry
- Danger zone + weak network
- Offline in high-risk area
- Long route deviation
- No user response after warning
- Repeated caution events

2. Live User Map

Show:

- User marker
- Last known location
- Current route
- Zone overlays
- Nearby help locations
- Route deviation
- Last update time

3. SOS Case Detail

Show:

- User name
- Phone

- Trip ID
- Emergency contacts
- Current/last known location
- Location timestamp
- Network state
- Risk reasons
- Route history
- Admin notes
- Action timeline
- Case status

4. Zone Management

Admin can:

- Create green/yellow/red zones
- Upload/import zone boundary polygons
- Add source/verification details
- Set expiry/review date
- Disable outdated zones
- Add road-closure/temporary restriction notices

5. Analytics

Show:

- Total active trips
- Total SOS events
- Most common caution areas
- Network-loss areas
- Route deviation heatmap
- Alert resolution time
- False-positive alert rate
- Most visited destinations

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20. SOS Workflow

User Side

User taps SOS
|

v
5-second countdown appears
|
+---- User cancels → no SOS event
|
v
SOS is confirmed
|
v
Capture current GPS + last known points
|
v
Create SOS alert in backend
|
v
Notify admin dashboard
|
v
Notify emergency contacts
|
v
User sees emergency support status

Admin Side

SOS appears at top of alert queue
|
v
Admin opens user and trip details
|
v
Admin acknowledges case
|
v
Admin contacts user if policy allows
|
v
Admin contacts emergency contact if required
|
v
Admin records action and notes
|
v
Case is resolved or escalated

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21. Privacy and Security Requirements

Privacy Rules

- Ask permission before collecting location.
- Location tracking should run only during an active trip.
- User must be able to pause or stop tracking.
- Explain how long location data will be retained.
- Allow user to delete or request deletion of trip/location data.
- Do not share exact location with unauthorized users.
- Do not expose raw GPS history to AI unnecessarily.
- Keep emergency contact sharing limited to the consented SOS process.

Security Rules

- Use HTTPS everywhere.
- Hash passwords using Argon2 or bcrypt.
- Keep API keys on server side only.
- Restrict Google Maps/Mapbox API keys by domain/IP/API.
- Use role-based access control.
- Use MFA for admin accounts.
- Add audit logs for:
    - Admin location access
    - Zone changes
    - SOS actions
    - User data export/deletion
- Add rate limiting to prevent abuse.
- Validate every API request.
- Use secure cookies or JWT refresh-token rotation.
- Encrypt highly sensitive information where feasible.
- Run security testing before launch.

---

22. Development Roadmap

Phase 0: Planning and Design

Build:

- Brand name and visual identity
- User flow diagrams
- Wireframes
- UI design system
- Database schema
- API documentation
- Privacy consent text
- SOS policy
- Pilot destination/city selection

Phase 1: Basic MVP

Build:

- Landing page
- Login/register
- User profile
- Emergency contacts
- Home dashboard
- Make a Trip wizard
- Unique Trip ID
- Basic itinerary screen
- Destination themes
- Basic admin login

Phase 2: Maps and Routes

Build:

- Destination map
- Geocoding
- Route display
- ETA
- Traffic-aware routes
- Alternate routes
- Nearby hospitals, police, fuel stations
- Route save/history

Phase 3: Safety MVP

Build:

- Tracking consent flow
- Start/stop trip tracking
- GPS location event storage
- Last known location
- Basic safety states
- Predefined green/yellow/red demo zones
- Basic geofence detection
- Safety status card
- User alerts

Phase 4: SOS System

Build:

- SOS button
- Countdown/cancel
- SOS backend event
- Admin SOS queue
- Emergency-contact notification
- Case acknowledgement
- Notes and audit timeline
- Sandbox testing

Phase 5: AI Features

Build:

- Hinglish travel chatbot
- Natural language trip form extraction
- AI itinerary generator
- Packing checklist generator
- Budget suggestion
- Safety alert explanation

- Admin incident summary
- Strict JSON schema validation

Phase 6: Advanced Intelligence

Build:

- Route-deviation scoring
- AI anomaly prioritization
- Dynamic risk scoring
- Personalized safety recommendations
- Multi-language support
- Destination knowledge base
- User feedback loop
- Analytics dashboard

---

23. Antigravity Ready Build Prompt

Copy this complete prompt into Antigravity:

Build a responsive, mobile-first web application named “SafeTrip AI”.

Product purpose:
SafeTrip AI helps users plan personalized trips and provides consent-based travel safety support during active trips using live location, geofenced safety zones, route awareness, alerts, and an SOS workflow.

Technology:
Use Next.js, React, TypeScript, Tailwind CSS, shadcn/ui, React Hook Form, Zod, TanStack Query, and Framer Motion.

Architecture:
Create a clean frontend architecture with typed service interfaces and mock data. Keep the app ready to connect later with a NestJS or FastAPI backend, PostgreSQL/PostGIS database, WebSockets, Google Maps/Mapbox, push notifications, and an AI API.

Brand:
Name: SafeTrip AI
Tagline: Plan Smart. Travel Safer.

Style: Modern premium travel app, calm, trustworthy, safety-first.
Main colors:
- Deep Teal: #0D617A
- Navy: #103B53
- Safe Green: #1F9D72
- Caution Amber: #F0A21A
- Danger Red: #D84343
- Background: #F7FAFB

Accessibility:
Use large touch targets, clear labels, strong contrast, keyboard support, visible focus styles, and never rely on color alone for safety status.

Pages:
1. Landing Page
2. Register Page
3. Login Page
4. Home Dashboard
5. Create Trip Wizard
6. AI Trip Plan Result Page
7. Active Trip Safety Dashboard
8. Live Map Page
9. SOS Flow/Modal
10. User Profile and Privacy Settings
11. Admin Login
12. Admin Alert Console
13. Admin SOS Case Detail
14. Admin Zone Management

Landing Page:
- Full-screen destination image/theme background.
- Logo and tagline.
- Login and Register CTA buttons.
- Small copy: “Personalized travel planning and active-trip safety support.”

Register:
- Name, email, phone, password or OTP, emergency contact.
- Consent checkboxes for terms, privacy, location tracking during active trip, SOS emergency contact sharing.
- Do not ask for device location permission here.

Home Dashboard:
- Greeting: “Hello, {name}! Ready for your next safe journey?”
- Cards: Total Trips, Active Trip, Estimated Budget, Saved Destinations.

- Tourist gallery cards.
- Main CTA: “Make a Trip”.
- Navigation to My Trips, Safety Dashboard, Profile and Emergency Contacts.

Trip Wizard:
Ask for:
- Starting location
- Destination
- Start and end date
- Number of days
- Budget
- Number of travelers
- Travel interests
- Transport preference
- Stay preference
- Food preference
- Special requirements

Trip Plan Result:
- Generate unique Trip ID.
- Show destination theme.
- Day-wise itinerary.
- Estimated budget split.
- Packing checklist.
- Things to explore.
- Hotel/stay suggestion cards.
- Three route options: Fastest, Lower-Risk, Scenic.
- Route table with ETA, traffic, network coverage and recommendation.
- CTA: “Start Trip Safety Mode”.

Tracking Consent Modal:
Before safety tracking begins, explain:
- What location is collected.
- Why it is collected.
- That tracking is active only during trip safety mode.
- That user can pause or stop tracking.
- That SOS shares last known/current location with authorized safety team and emergency contacts.
- Add buttons: “Allow and Start Tracking” and “Not Now”.

Active Trip Safety Dashboard:
Top bar:
- Welcome, {name}
- Current Trip ID

- Overall safety badge
- Notification icon
- Profile icon

Create 4 main tabs:
1. Profile
2. Safety Status
3. Live Map
4. SOS

Profile Tab:
- User ID
- Trip ID
- Name
- Email
- Phone
- Emergency Contact
- Destination
- Travel dates
- Tracking status

Safety Status Tab:
- Overall Status: Safe, Caution, Danger or Offline/Last Seen.
- Area Zone
- Network State
- GPS last update
- GPS accuracy
- Route deviation
- Risk explanation
- Recommended action
- Tracking pause/stop controls

Live Map:
- Use mock map if API key is not available.
- Display user marker.
- Display planned route polyline.
- Display green, yellow and red zone overlays.
- Show nearby hospital, police station, fuel station and hotel markers.
- Show last known location if user is offline.
- Show route alternatives.
- Show “Last updated” timestamp.

Safety Risk Engine:
Build a transparent mocked rule-based risk engine.

Inputs:
- Area risk
- Route deviation
- Time risk
- Network risk
- GPS freshness and accuracy
- Traffic/weather/road risk

Formula:
Risk Score =
0.35 * AreaRisk +
0.20 * RouteDeviation +
0.15 * TimeRisk +
0.10 * NetworkRisk +
0.10 * GPSQualityRisk +
0.10 * ExternalEventRisk

Rules:
- Safe: score 0–34
- Caution: score 35–64
- Danger: score 65–100
- Offline/Last Seen: stale location update; not automatically danger.
- Never mark danger based only on weak network.
- Every status must show reason codes and user-friendly explanation.

SOS:
- Build a large red SOS button.
- On click, open a full-screen confirmation panel.
- Start a 5-second countdown.
- User can cancel.
- After countdown, create a mocked SOS case.
- Show current or last known location.
- Show message: “Your SOS alert has been sent to the SafeTrip safety team and your selected emergency contacts.”
- Do not claim police/ambulance dispatch.
- Include Call Emergency Contact and Share Location options.

AI Travel Assistant:
- Add a chatbot component supporting English, Hindi and Hinglish.
- Use mock structured JSON responses for now.
- Use AI for:
  - Natural language trip extraction
  - Itinerary generation
  - Packing checklist

  - Budget suggestion
  - Route explanation
  - Safety alert explanation
  - Admin incident summary
- Design typed interfaces for future API integration.
- AI must not autonomously trigger SOS or override deterministic safety rules.

Admin Dashboard:
- Alert priority queue.
- Tabs: All Alerts, SOS Alerts, Danger Zones, Offline Users, Resolved.
- User/trip side panel.
- Map with current/last known location.
- Timeline of location and risk events.
- SOS case actions: Acknowledge, Contacted, Escalated, Resolved.
- Admin notes and audit log.
- Zone management page with create/edit/delete zone forms and map polygon mock interface.

Data Models:
- User
- UserConsent
- EmergencyContact
- Trip
- Route
- SafetyZone
- LocationEvent
- RiskEvent
- SOSAlert
- AdminAction
- Notification

Privacy:
- Add tracking on/off toggle.
- Add end trip flow.
- Add consent history section.
- Add delete data button.
- Display retention policy text.
- Never show exact user location outside authorized user/admin role.

Deliverables:
- Polished responsive UI.
- Complete clickable flow.
- Reusable components.
- TypeScript types.
- Mock API/service layer.

- Realistic mock data.
- Empty, loading and error states.
- README with setup instructions.
- README section explaining future backend, map, WebSocket and AI integration.

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24. Final Build Priority

Do not try to build everything at once.

Build in this order:

1. Login/Register
2. Trip Planner
3. Home Dashboard
4. Trip ID and Itinerary
5. Map and Route UI
6. Consent-Based GPS Tracking
7. Zone Detection
8. Safety Status
9. SOS
10. Admin Dashboard
11. AI Travel Assistant
12. Advanced AI/ML Risk Detection

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25. Final Product Statement

SafeTrip AI is not just a travel website.

It is a:

> **Travel planning platform + AI travel companion + location-aware safety system + SOS support dashboard.**

The strongest implementation model is:

Verified Maps and Zone Data
+
Rule-Based Safety Engine
+

Live GPS and Route Context
+
Human Admin Oversight
+
AI for Personalization, Explanation and Summaries

Do not build the system as:

AI prediction only

The AI should make the product smarter, but verified data, deterministic safety logic, user consent, and human review should keep it trustworthy.

P